

ACTIVITY 14 — FACILITATOR GUIDE**Risk Register, RBS, EMV and Decision Tree**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO4	3 - Plan the Project

Purpose of this activity

By the end of this activity learners will be able to build a risk breakdown structure and use it to identify risks systematically rather than by brainstorm alone.

What learners produce

risk register, rbs, pi matrix — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Build the risk breakdown structure. Create artifacts/14-rbs.md. The RBS is a hierarchy of risk SOURCES. Its purpose is systematic identification: working through categories catches risks that free brainstorming misses, because brainstorming surfaces what is memorable rather than what is likely.	Keep role-holders in role; learners drift to consensus. Push for the concern behind the stated position.
2	Trace risks from Lab 02 factors and Lab 06 assumptions. Risk identification is not invention. Most risks on a well-run project are already sitting in earlier artifacts, unlabelled.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
3	Write risk statements in cause-event-effect form	Ask what the number would have to be before the decision flips. That is the exam's favourite

		question.
4	Build the risk register. Create artifacts/14-risk-register.md. Probability is given as a decimal and as a 1-5 score. Impact is scored 1-5, with the cost impact in SGD and the schedule impact in days.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
5	Plot the probability/impact matrix. Create artifacts/14-pi-matrix.md. Score = probability score x impact score, both on 1-5.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
6	Apply all ten response strategies. A real Contoso example of each of the ten:	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Compute expected monetary value. Sign convention: threat impacts are shown as positive cost figures and summed as exposure; opportunity impacts are shown as negative because they reduce exposure.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
8	Build and solve a decision tree. Create artifacts/14-emv-analysis.md. A decision tree evaluates a choice under uncertainty by computing the EMV of each branch and selecting the best.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
9	Qualitative versus quantitative, and when a risk becomes an issue	Ask for one item the group nearly dropped and what made them keep it.
10	Answer the exam-style scenarios	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.

Suggested timing

Minutes	Phase
9 min	Brief the scenario, confirm the inputs and assign roles
33 min	Work the first half of the steps individually, then pool as a group
34 min	Complete the remaining steps and finalise the deliverable
14 min	Group share-back, facilitator debrief and exam-style linkage

Total: 90 minutes

Discussion & decision prompts

1. Build a risk breakdown structure and use it to identify risks systematically rather than by brainstorm alone — how did your group decide, and what would change that decision?
2. Trace risks from the Lab 02 environment scan and the Lab 06 assumption log, since every assumption is a latent risk — how did your group decide, and what would change that decision?
3. Write risk statements in cause-event-effect form so that the cause can be attacked and the effect can be priced — how did your group decide, and what would change that decision?
4. Every risk statement contains a cause that is a present fact, an uncertain event, and an effect on a named project objective. If any statement is a single noun phrase, it cannot be responded to or priced — can every member of your group defend it?
5. Where would risk register, rbs, emv and decision tree have saved you time or money on a real project you have worked on?

What good looks like

- Every risk statement contains a cause that is a present fact, an uncertain event, and an effect on a named project objective. If any statement is a single noun phrase, it cannot be responded to or priced.

- Your register contains at least two opportunities with positive EMV. A register of threats only is half a risk register.
- Every assumption in the Lab 06 log has a corresponding risk, and you can show the conversion for A-05.
- Every risk has a trigger that someone could actually observe. "Monitor closely" is not a trigger.
- Your P/I scores are consistent with your decimal probabilities - a risk at 0.45 should not score 2 on a 1-5 probability scale.
- Your residual net EMV is approximately SGD 26,600, reconciling to the SGD 26,000 contingency with the SGD 600 variance stated rather than hidden.
- Your opportunity probabilities INCREASED after response while your threat probabilities decreased. If both moved the same way, you applied threat logic to opportunities.
- Your decision tree branch probabilities sum to exactly 1.0 at each chance node, and the decision node costs are included in the path totals.